

Data Engineering Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

1. Introduction to Data Engineering

- What is Data Engineering?
- Role and Responsibilities of a Data Engineer
- Data Engineer vs Data Analyst
- Data Engineer vs Data Scientist
- Data Engineer vs DBA
- Data Engineer vs Analytics Engineer
- Data Engineering Lifecycle
- Structured, Semi-Structured and Unstructured Data
- Batch, Streaming, Transactional and Analytical Data
- Modern Data Architecture

2. Linux Fundamentals for Data Engineers

- Linux File System
- pwd, ls, cd, mkdir, touch, cp, mv, rm
- cat, less, head, tail
- File Permissions
- chmod and chown
- grep, cut, sort, uniq, wc
- awk and sed
- Process Management
- Environment Variables
- Shell Scripting
- Variables, Conditions, Loops and Functions
- Automating Data Tasks

3. Python for Data Engineering

- Python Fundamentals
- Lists, Tuples, Sets and Dictionaries
- Functions, Modules and Packages
- File Processing
- CSV, JSON, XML, Parquet and Avro
- Exception Handling
- Object-Oriented Programming
- Database Connectivity
- Connection Management and Transactions
- REST APIs
- HTTP Methods
- Authentication
- Pagination and Rate Limits
- Large File Processing
- Chunk Processing
- Generators
- Logging
- Configuration Management

4. SQL for Data Engineering

- Database, Schema, Table, Row, Column and Data Types
- CREATE, ALTER, DROP and TRUNCATE
- INSERT, UPDATE, DELETE and MERGE
- SELECT, WHERE, DISTINCT, ORDER BY and LIMIT
- COUNT, SUM, AVG, MIN and MAX
- GROUP BY and HAVING
- INNER, LEFT, RIGHT, FULL, CROSS and SELF JOIN
- Subqueries and Correlated Subqueries
- CTEs and Recursive CTEs
- ROW_NUMBER, RANK, DENSE_RANK, LAG and LEAD
- SUM OVER, AVG OVER and PARTITION BY
- Views and Materialized Views
- Stored Procedures and User-Defined Functions
- ACID Transactions
- COMMIT, ROLLBACK and SAVEPOINT
- Isolation Levels
- Execution Plans
- Indexes
- Partitioning
- Query Optimization

5. Relational Database Systems

- Relational Database Fundamentals
- Keys, Constraints and Relationships
- Primary Key, Foreign Key, Unique, Not Null and Check
- First, Second and Third Normal Form
- BCNF
- Denormalization
- Clustered and Non-Clustered Indexes
- Composite and Covering Indexes
- Range, List and Hash Partitioning
- PostgreSQL
- MySQL
- SQL Server Concepts
- Managed Cloud Databases

6. NoSQL Databases

- Why NoSQL?
- CAP Theorem
- Eventual Consistency
- Horizontal Scaling
- Document Databases
- Key-Value Databases
- Column-Family Databases
- Graph Database Concepts
- MongoDB
- Collections and Documents
- BSON
- CRUD and Aggregation Pipeline
- Redis

- Caching and TTL
- Cassandra Concepts
- Partition and Clustering Keys
- SQL vs NoSQL Selection

7. Data Modeling

- Conceptual, Logical and Physical Modeling
- Entity Relationship Modeling
- Entities, Attributes, Relationships and Cardinality
- OLTP Modeling
- OLAP Modeling
- Dimensional Modeling
- Star Schema
- Snowflake Schema
- Fact Constellation
- Transaction Facts
- Periodic Snapshot Facts
- Accumulating Snapshot Facts
- Conformed and Role-Playing Dimensions
- Degenerate and Junk Dimensions
- Slowly Changing Dimensions Type 0, 1, 2, 3 and 4
- Natural vs Surrogate Keys

8. Data Warehousing

- Operational vs Analytical Systems
- OLTP vs OLAP
- Enterprise Data Warehouse
- Data Mart
- Source, Staging, Integration, Warehouse and Presentation Layers
- Grain
- Facts, Dimensions and Measures
- Aggregations
- Full Load
- Incremental Load
- Delta Load
- Upsert and Merge
- Historical Data
- Effective Dates and Current Flags
- Snowflake
- Amazon Redshift
- Google BigQuery
- Azure Synapse Analytics

9. ETL and ELT

- Extract, Transform and Load
- Extract, Load and Transform
- ETL vs ELT
- Database, API, File, Object Storage and SaaS Sources
- Cleaning, Standardization, Deduplication and Filtering
- Joins, Aggregations and Business Rules
- Append, Overwrite, Merge and Upsert

- Full Refresh
- Incremental Load
- Watermark-Based Loading
- Change Data Capture
- Idempotent Pipelines

10. Data Pipeline Engineering

- Sources, Sinks, Transformations and Dependencies
- Batch Pipelines
- Streaming Pipelines
- Reliability
- Scalability
- Fault Tolerance
- Idempotency
- Reprocessing and Backfills
- Modular and Reusable Pipelines
- Configuration-Driven Pipelines
- Parameterized Pipelines
- Retries
- Dead-Letter Handling
- Logging
- Alerting
- Failure Recovery

11. Apache Airflow

- Workflow Orchestration
- DAG, Task, Operator, Scheduler and Executor
- Creating DAGs
- Scheduling and Dependencies
- PythonOperator, BashOperator and SQL Operators
- Sensors
- Upstream and Downstream Dependencies
- Branching
- Trigger Rules
- XCom
- Dynamic Task Mapping
- Parameterized DAGs
- Backfill
- Catchup
- Retry
- SLA
- Monitoring
- Secrets Management
- DAG Testing
- Production Best Practices

12. dbt – Data Build Tool

- Analytics Engineering
- Models, Sources and Seeds
- Views, Tables and Incremental Models
- Ephemeral Models

- Source Freshness
- Unique, Not Null, Accepted Values and Relationship Tests
- Custom Tests
- Documentation
- Lineage Graph
- Jinja
- Variables, Conditions and Loops
- Macros
- Incremental Processing
- Snapshots
- dbt Project Structure

13. Change Data Capture

- Insert, Update and Delete Capture
- Timestamp-Based CDC
- Trigger-Based CDC
- Log-Based CDC
- Database Transaction Logs
- Source Database, Connector, Message Broker and Destination
- Debezium Concepts
- Kafka Connect
- Cloud CDC Services
- Replication and Event-Driven Use Cases

14. Data Formats

- CSV
- JSON
- Avro
- Parquet
- ORC
- Row vs Columnar Storage
- Compression
- Predicate Pushdown
- Schema Evolution
- Choosing File Formats

15. Data Compression and Serialization

- Lossless Compression
- Compression Ratio
- CPU vs Storage Trade-Off
- gzip
- Snappy
- Zstandard
- JSON and Avro Serialization
- Backward and Forward Compatibility
- Schema Registry Concepts

16. Distributed Computing Fundamentals

- Nodes and Clusters
- Distributed Storage
- Distributed Processing

- Vertical vs Horizontal Scaling
- Network and Partial Failures
- Latency
- Replication
- Consistency
- Hash and Range Partitioning
- Fault Tolerance
- Retry and Checkpointing

17. Hadoop Ecosystem

- Hadoop Fundamentals
- HDFS
- NameNode
- DataNode
- Blocks and Replication
- MapReduce
- Mapper, Reducer, Shuffle and Sort
- YARN
- Resource Manager
- Node Manager
- Application Master
- Hive
- HBase
- Sqoop Concepts
- Hadoop Legacy Architecture

18. Apache Spark

- Spark Fundamentals
- Distributed and In-Memory Processing
- Driver and Executors
- Cluster Manager
- Jobs, Stages and Tasks
- SparkSession
- RDDs
- Transformations and Actions
- DataFrames
- Schema
- Selection, Filtering and Sorting
- withColumn, drop, rename, cast and when
- Aggregations
- Joins
- Broadcast Join
- Spark SQL
- Temporary Views
- Window Functions

19. PySpark

- PySpark Environment
- SparkSession
- Reading CSV, JSON, Parquet, ORC and Database Data
- Writing Data

- Save Modes
- Partitioned Output
- select, filter, where, withColumn, groupBy and join
- Python UDF
- Pandas UDF
- Null Handling
- Date, Timestamp and String Functions

20. Spark Performance Optimization

- Lazy Evaluation
- Execution Plans
- repartition and coalesce
- Partition Size
- Data Skew
- Salting
- Caching and persist
- Storage Levels
- Broadcast Join
- Shuffle Join
- Sort-Merge Join
- Catalyst Optimizer
- Predicate Pushdown
- Column Pruning
- Adaptive Query Execution
- Skew Handling

21. Apache Kafka

- Event Streaming
- Message Broker
- Event-Driven Architecture
- Producer, Consumer, Broker, Topic, Partition and Offset
- Producer Keys and Partition Selection
- Acknowledgements
- Consumer Groups
- Parallel Consumption
- Topic Partitioning
- Leader and Followers
- Replication Factor
- At-Most-Once
- At-Least-Once
- Exactly-Once Concepts
- Kafka Connect
- Source and Sink Connectors
- Schema Registry Concepts

22. Stream Processing

- Batch vs Streaming
- Event Time and Processing Time
- Windows
- Watermarks
- Late Arriving Events

- Stateful Processing
- Spark Structured Streaming
- Streaming Sources and Sinks
- Triggers
- Tumbling, Sliding and Session Windows
- Stateful Aggregation
- Deduplication
- Kafka + Spark Architectures

23. Data Lakes

- Data Lake Fundamentals
- Raw/Bronze Layer
- Cleaned/Silver Layer
- Curated/Gold Layer
- Amazon S3
- Azure Data Lake Storage
- Google Cloud Storage
- Data Swamp Risks
- Schema Management
- Governance
- Date and Region Partitioning
- Partition Pruning

24. Lakehouse Architecture

- Data Lake + Data Warehouse
- ACID Transactions
- Schema Enforcement
- Schema Evolution
- Versioning
- Time Travel
- Bronze, Silver and Gold Layers
- Delta Lake
- Apache Iceberg
- Apache Hudi
- MERGE and Upsert

25. Databricks

- Workspace
- Compute
- Notebooks
- Jobs
- Spark Clusters
- PySpark
- Spark SQL
- Delta Tables
- MERGE and Time Travel
- Medallion Architecture
- Databricks Workflows
- Tasks and Dependencies
- Unity Catalog Concepts
- Catalogs, Schemas and Tables

- Access Control and Lineage

26. Snowflake

- Snowflake Architecture
- Storage, Compute and Cloud Services Layers
- Virtual Warehouses
- Auto Suspend and Auto Resume
- Stages
- COPY INTO
- Databases, Schemas, Tables and Views
- Streams and Tasks
- Time Travel
- Micro-Partitions
- Clustering
- Query Profile
- Result Cache

27. Cloud Data Engineering Fundamentals

- Compute
- Storage
- Databases
- Networking
- Identity and Access Management
- Object, Block and File Storage
- Managed Relational and NoSQL Databases
- Managed Data Warehouses
- Serverless Data Processing
- Cloud Security
- Secrets
- Encryption

28. Microsoft Azure Data Engineering

- Azure Blob Storage
- Azure Data Lake Storage Gen2
- Azure Data Factory
- Pipelines, Activities, Linked Services and Datasets
- Integration Runtime
- Triggers
- Copy Activity
- ForEach, If Condition and Lookup
- Execute Pipeline
- Azure Databricks
- Azure Synapse Analytics
- Dedicated and Serverless SQL
- Spark Pools
- Azure Event Hubs
- Azure Functions

29. AWS Data Engineering Overview

- Amazon S3
- AWS Glue

- Data Catalog
- Crawlers and ETL Jobs
- Amazon Redshift
- Amazon Kinesis
- Amazon EMR
- AWS Lambda

30. Google Cloud Data Engineering Overview

- Google Cloud Storage
- BigQuery
- Partitioning and Clustering
- Dataflow
- Pub/Sub
- Dataproc

31. Data Quality Engineering

- Accuracy, Completeness, Consistency, Validity, Uniqueness and Timeliness
- Schema Validation
- Type and Range Validation
- Null Checks
- Referential Integrity
- Source, Transformation and Destination Tests
- Business Rules
- Duplicate and Freshness Checks
- dbt Tests
- Great Expectations Concepts

32. Data Observability

- Data Health
- Pipeline Success and Failure Rates
- Runtime and Throughput
- Freshness
- Volume
- Schema
- Distribution
- Data Drift
- Alerting
- SLA Breaches
- Root Cause Analysis
- Logs and Lineage

33. Data Governance

- Ownership
- Stewardship
- Policies and Standards
- Data Catalog
- Dataset Discovery
- Metadata
- Dataset-Level and Column-Level Lineage
- Technical, Business and Operational Metadata
- Data Classification

- Retention Policies
- Archival and Deletion

34. Data Security

- Authentication
- Service Accounts and Managed Identities
- Authorization
- Role-Based Access Control
- Least Privilege
- Encryption at Rest
- Encryption in Transit
- Secrets Management
- Static and Dynamic Data Masking
- PII Identification and Protection

35. Data Privacy and Compliance

- Personal and Sensitive Data
- Consent
- Data Minimization
- Masking
- Encryption
- Tokenization
- Anonymization
- GDPR Concepts
- Data Residency
- Retention Requirements
- Auditability

36. Git and Version Control

- Repository
- Clone, Add, Commit, Push and Pull
- Branching
- Merge and Rebase Concepts
- GitHub
- Pull Requests
- Issues
- Code Reviews
- Data Engineering Repository Structure

37. Docker for Data Engineering

- Containers and Images
- Docker Engine
- Dockerfile
- FROM, COPY, RUN, WORKDIR, CMD and ENTRYPOINT
- Volumes
- Networking
- Docker Compose
- Multi-Service Applications
- Local Kafka, Airflow, Spark and Database Environments

38. CI/CD and DataOps

- DataOps Fundamentals
- Continuous Integration
- Continuous Delivery
- Continuous Deployment
- Unit Tests
- Integration Tests
- Data Tests
- Dev/Test/Stage/Production Environments
- GitHub Actions
- Environment Variables
- Configuration Files
- Secrets

39. Infrastructure as Code

- Declarative Infrastructure
- Terraform Fundamentals
- Providers
- Resources
- Variables
- Outputs
- State
- Storage, Database, Compute and Network Provisioning
- Environment Management

40. Data Architecture Patterns

- Batch Architecture
- Lambda Architecture
- Kappa Architecture
- Lakehouse Architecture
- Medallion Architecture
- Event-Driven Architecture

41. Data Partitioning Strategies

- Why Partition Data?
- Partition Pruning
- Time-Based Partitioning
- Hash Partitioning
- Range Partitioning
- Choosing Partition Keys
- Cardinality and Query Patterns
- Too Many Small Partitions
- Data Skew

42. Incremental Data Processing

- New, Updated and Deleted Records
- Timestamp Watermarks
- Incrementing IDs
- Merge and Upsert
- Idempotency

- Late Arriving Facts and Dimensions
- Reprocessing
- Historical Backfills

43. Data Lineage and Metadata

- Technical, Business and Operational Metadata
- Dataset-Level Lineage
- Column-Level Lineage
- Pipeline Lineage
- Data Catalog
- Ownership and Documentation
- Impact Analysis
- Downstream Dependencies

44. Performance Optimization

- Database Indexing and Query Plans
- Partitioning and Materialized Views
- Parquet and Compression
- File Size Optimization
- Spark Partitioning and Caching
- Broadcast Joins
- Skew Handling
- Warehouse Clustering
- Compute Sizing
- Parallelism
- Incremental Processing
- Reduced Data Movement
- Pushdown Processing

45. Cost Optimization

- Storage, Compute and Network Cost
- Compression
- Lifecycle Policies
- Tiered Storage
- Autoscaling
- Right-Sizing
- Cluster Shutdown
- Partition Pruning
- Incremental Processing
- Avoiding Unnecessary Reprocessing

46. Data Engineering System Design

- Requirement Gathering
- Data Volume, Velocity and Variety
- Latency and Availability Requirements
- Storage Selection
- Relational vs NoSQL vs Lake vs Warehouse vs Lakehouse
- Batch vs Streaming
- Hybrid Processing
- Pipeline Design
- Scalability

- Reliability
- Retries
- Checkpointing
- Idempotency
- Disaster Recovery
- E-Commerce, Financial, IoT and Log Analytics Scenarios

47. Mini Projects

- Batch ETL Pipeline
- Data Warehouse
- Airflow Data Pipeline
- dbt Analytics Engineering Project
- PySpark Data Pipeline
- Real-Time Kafka Pipeline
- CDC Pipeline
- Cloud Data Lake
- Lakehouse Project

48. Capstone Project

- Source Systems
- Relational Database, API, Files and Event Streams
- Batch, API, CDC and Kafka Ingestion
- Raw Data Lake
- Bronze/Silver/Gold Layers
- PySpark Processing
- Delta Lake
- Data Warehouse
- Fact and Dimension Modeling
- SCD Type 2
- dbt Models and Tests
- Airflow Orchestration
- Kafka Streaming
- Data Quality
- Observability
- Governance and Lineage
- Git, Docker and CI/CD
- Cloud Deployment

49. Interview Preparation

- Advanced SQL
- Databases
- Indexing, Partitioning and Transactions
- Data Modeling
- ETL/ELT
- CDC and Idempotency
- Spark and PySpark
- Kafka
- Airflow
- Data Lakes and Lakehouses
- Cloud
- Pipeline and Platform System Design

- Mock Interviews
- Python
- SQL
- Shell Scripting
- Linux
- PostgreSQL
- MySQL
- MongoDB
- Redis
- Cassandra concepts
- Apache Airflow
- dbt
- Apache Spark
- PySpark
- Apache Kafka
- Kafka Connect
- Debezium concepts
- Spark Structured Streaming
- Hadoop/HDFS concepts
- CSV
- JSON
- Parquet
- Avro
- ORC
- Delta Lake
- Apache Iceberg concepts
- Apache Hudi concepts
- Databricks
- Snowflake
- Azure Data Factory
- Azure Data Lake Storage
- Azure Synapse Analytics
- Azure Event Hubs
- AWS S3
- AWS Glue
- Amazon Redshift
- Amazon Kinesis
- Google BigQuery
- Google Cloud Storage
- Great Expectations concepts
- Docker
- Docker Compose
- Terraform
- Git
- GitHub
- GitHub Actions

Enquire or book a free demo session. Archer Infotech, Flat No. 12, Divyadarshan Housing Society, behind Kothrud Bus Stand Road, Kothrud, Pune 411038. Phone +91 9850 678451. Email info@archerinfotech.in. Web archerinfotech.in.